

# SNAPDRAGON® AI LAB BUILD & PRESENT CHALLENGE

## SmartQC

On-device visual quality inspection, built and optimized for Snapdragon.

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## THE PROBLEM

# Quality inspection is slow, inconsistent, or expensive

1

### Manual inspection

Depends on human attention span — fatigue and inconsistency creep in over a shift.

2

### Cloud-based AI tools

Solve accuracy, but add latency, an internet dependency, and data-privacy exposure.

3

### Industrial inspection hardware

Accurate and fast, but priced far out of reach for small food producers and workshops.

There is a need for a fast, private, self-contained inspection tool that runs entirely on the device performing the inspection.

## THE SOLUTION

# SmartQC: factory-grade inspection, without the factory-grade price tag

### WHO IT'S FOR

Small bakeries, food producers, and workshops that need reliable visual QC but can't justify a dedicated inspection line.

### WHAT IT DOES

Classifies product quality from a webcam or uploaded image, entirely on-device — no cloud calls, no internet dependency at inference time.

### WHY IT MATTERS

Runs on a Snapdragon laptop the business may already own — zero extra hardware cost.

### VALUE PROPOSITION

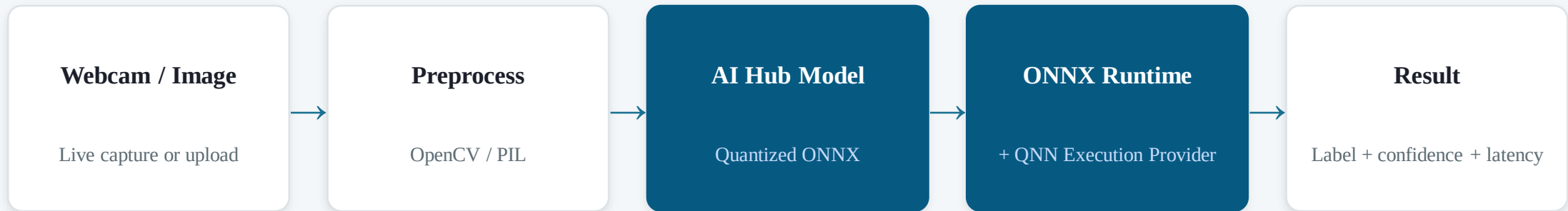
**“Factory-grade visual inspection,  
without the factory-grade price tag.”**

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Built by extending an earlier ESP32-CAM + MobileNetV2 bread-quality system into a fully software-based Windows app for Snapdragon-powered laptops.

## HOW IT WORKS

# Architecture



The optimized model and ONNX Runtime + QNN Execution Provider path are what make this pipeline Snapdragon-NPU-ready.

**App features:** live webcam / upload inspection • batch folder inspection + CSV export • confidence threshold flagging • local SQLite history log • rolling defect-rate trend chart • fully offline at inference time

WHY ON-DEVICE, WHY SNAPDRAGON

# Built for where inspection actually happens

## Cloud-based inspection

- ✗ Needs constant internet
- ✗ Per-image / API cost
- ✗ Data leaves the device
- ✗ Latency varies with connection

## SmartQC on Snapdragon

- ✓ Fully offline at inference
- ✓ Zero marginal cost per scan
- ✓ All data stays on-device
- ✓ NPU-optimized, low-latency

Snapdragon's on-device NPU + long battery life means SmartQC can run all day on a shop floor — unplugged, private, and instant.

## MODEL TRAINING

# MobileNetV2, fine-tuned to 99.2% validation accuracy

99.2%

Best validation accuracy

649

Total labeled images (Fresh / Mold bread)

12

Training epochs

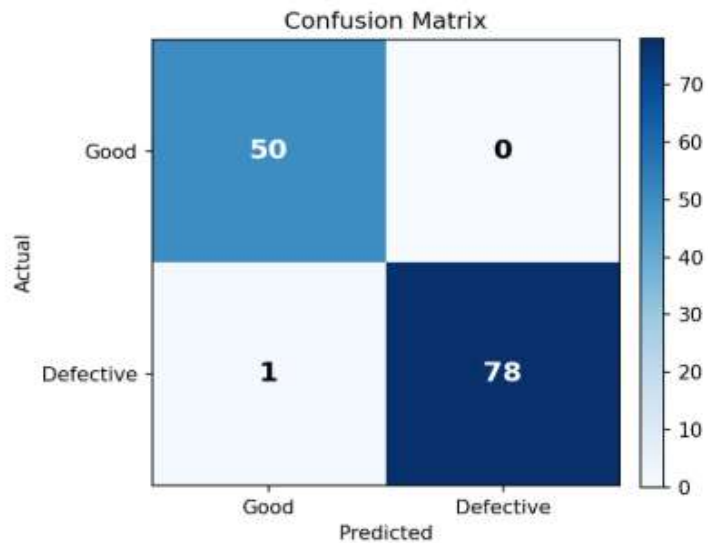
```
● (venv) PS C:\Users\samay\OneDrive\Desktop\SmartQC\app> cd ..
● (venv) PS C:\Users\samay\OneDrive\Desktop\SmartQC> python train.py
Found classes: ['Fresh Bread(Good Bread) Images', 'Mold Bread(Bad Bread) Images']
Total images: 649
Train: 520 Val: 129
Epoch 1/12 train_loss=0.2621 train_acc=0.902 val_acc=0.984
Epoch 2/12 train_loss=0.1234 train_acc=0.956 val_acc=0.992
Epoch 3/12 train_loss=0.0647 train_acc=0.977 val_acc=0.992
Epoch 4/12 train_loss=0.0595 train_acc=0.981 val_acc=0.984
Epoch 5/12 train_loss=0.0374 train_acc=0.992 val_acc=0.984
Epoch 6/12 train_loss=0.0208 train_acc=0.994 val_acc=0.984
Epoch 7/12 train_loss=0.0169 train_acc=0.994 val_acc=0.977
Epoch 8/12 train_loss=0.0135 train_acc=0.996 val_acc=0.984
Epoch 9/12 train_loss=0.0538 train_acc=0.992 val_acc=0.984
Epoch 10/12 train_loss=0.0698 train_acc=0.977 val_acc=0.984
Epoch 11/12 train_loss=0.0220 train_acc=0.990 val_acc=0.969
Epoch 12/12 train_loss=0.0202 train_acc=0.996 val_acc=0.984

Best validation accuracy: 0.992
```

Fine-tuned on a Fresh Bread (Good) / Mold Bread (Defective) dataset using transfer learning from ImageNet-pretrained MobileNetV2 — the same architecture family used in the original bread-quality-indicator project this builds on.

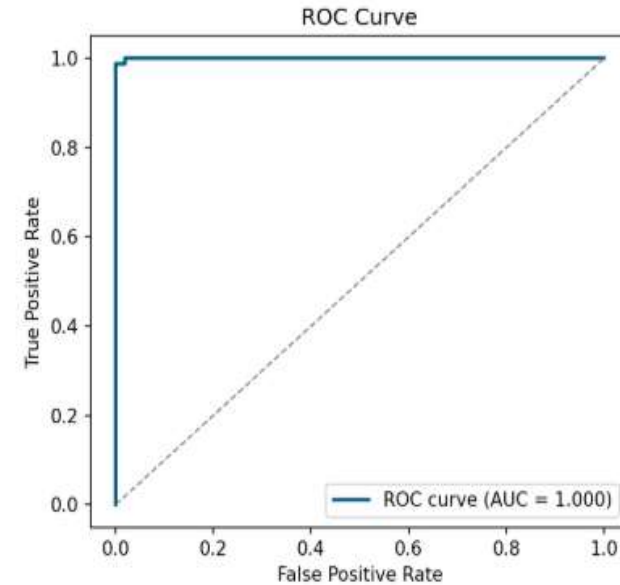
## MODEL EVALUATION

# Beyond accuracy: confusion matrix, precision/recall, ROC-AUC



**129**

Validation images



**1**

Total misclassifications

**0.9997**

ROC-AUC

# Compiled and optimized for the Snapdragon X Elite

COMPILE

QUANTIZE

LINK

PROFILE

INFERENCE

Creator: samaythakur999@gmail.com (me)

Rows per page: 10

1-3 of 3

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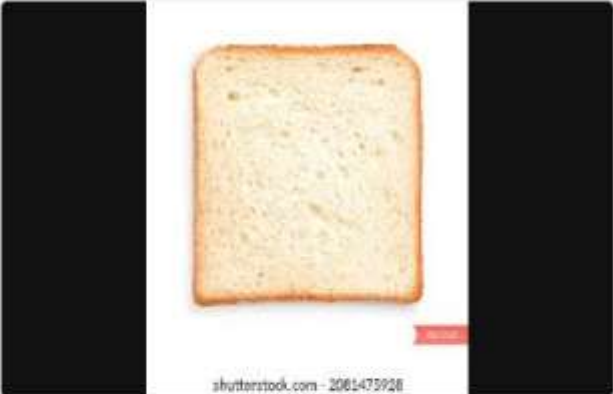
| Job Name           | Job ID    | Status        | Target Device          | Creator                  | Submission Time       |
|--------------------|-----------|---------------|------------------------|--------------------------|-----------------------|
| smartqc_model.onnx | jpe782z15 | Results Ready | Snapdragon X Elite CRD | samaythakur999@gmail.com | 9/20/2026, 2:51:55 AM |

Compile job jpe782z15 • Target: Snapdragon X Elite CRD • Status: Results Ready

LIVE DEMO

Correctly classifies both good and defective samples

SmartQC — On-Device Visual Quality Inspection



shutterstock.com - 2061475928

Start Webcam Upload Image Backend: cpu (local) Run Inspection

Result: GOOD (100.0%)

Latency: 3.9 ms [CPU]

Run Timing Comparison (local CPU)

Note: this compares repeated local inference calls, not a validated NPU benchmark.  
The model is compiled/optimized for the Snapdragon NPU via Qualcomm AI Hub.  
(see the AI Hub job list in the README) — NPU execution provider deployment  
is the final integration step.

Run 1: 5.6 ms | Run 2: 7.7 ms (both local CPU — see note above)

| History             |               | Trend  |     |
|---------------------|---------------|--------|-----|
| 2026-09-19 22:13:36 | good (100.0%) | 3.9 ms | cpu |
| 2026-09-19 22:13:36 | good (100.0%) | 4.1 ms | cpu |
| 2026-09-19 22:13:36 | good (100.0%) | 4.6 ms | cpu |
| 2026-09-19 22:13:36 | good (100.0%) | 4.5 ms | cpu |
| 2026-09-19 22:13:36 | good (100.0%) | 4.1 ms | cpu |
| 2026-09-19 22:13:36 | good (100.0%) | 4.3 ms | cpu |
| 2026-09-19 22:13:35 | good (100.0%) | 4.6 ms | cpu |
| 2026-09-19 22:13:35 | good (100.0%) | 4.2 ms | cpu |
| 2026-09-19 22:13:35 | good (100.0%) | 4.4 ms | cpu |
| 2026-09-19 22:13:35 | good (100.0%) | 4.4 ms | cpu |
| 2026-09-19 22:13:35 | good (100.0%) | 4.5 ms | cpu |
| 2026-09-19 22:13:34 | good (100.0%) | 4.7 ms | cpu |
| 2026-09-19 22:13:34 | good (100.0%) | 5.8 ms | cpu |
| 2026-09-19 22:13:34 | good (100.0%) | 5.0 ms | cpu |
| 2026-09-19 22:13:34 | good (100.0%) | 4.3 ms | cpu |
| 2026-09-19 22:13:34 | good (100.0%) | 4.8 ms | cpu |
| 2026-09-19 22:13:34 | good (100.0%) | 5.2 ms | cpu |
| 2026-09-19 22:13:34 | good (100.0%) | 5.3 ms | cpu |
| 2026-09-19 22:13:33 | good (100.0%) | 5.5 ms | cpu |
| 2026-09-19 22:13:33 | good (100.0%) | 6.0 ms | cpu |
| 2026-09-19 22:13:33 | good (100.0%) | 4.8 ms | cpu |
| 2026-09-19 22:13:33 | good (100.0%) | 4.3 ms | cpu |
| 2026-09-19 22:13:33 | good (100.0%) | 5.0 ms | cpu |
| 2026-09-19 22:13:32 | good (100.0%) | 4.2 ms | cpu |
| 2026-09-19 22:13:32 | good (100.0%) | 5.7 ms | cpu |

GOOD — 100.0% confidence

SmartQC — On-Device Visual Quality Inspection



Start Webcam Upload Image Backend: cpu (local) Run Inspection

Result: DEFECTIVE (98.4%)

Latency: 4.3 ms [CPU]

Run Timing Comparison (local CPU)

Note: this compares repeated local inference calls, not a validated NPU benchmark.  
The model is compiled/optimized for the Snapdragon NPU via Qualcomm AI Hub.  
(see the AI Hub job list in the README) — NPU execution provider deployment  
is the final integration step.

Run 1: 4.5 ms | Run 2: 3.6 ms (both local CPU — see note above)

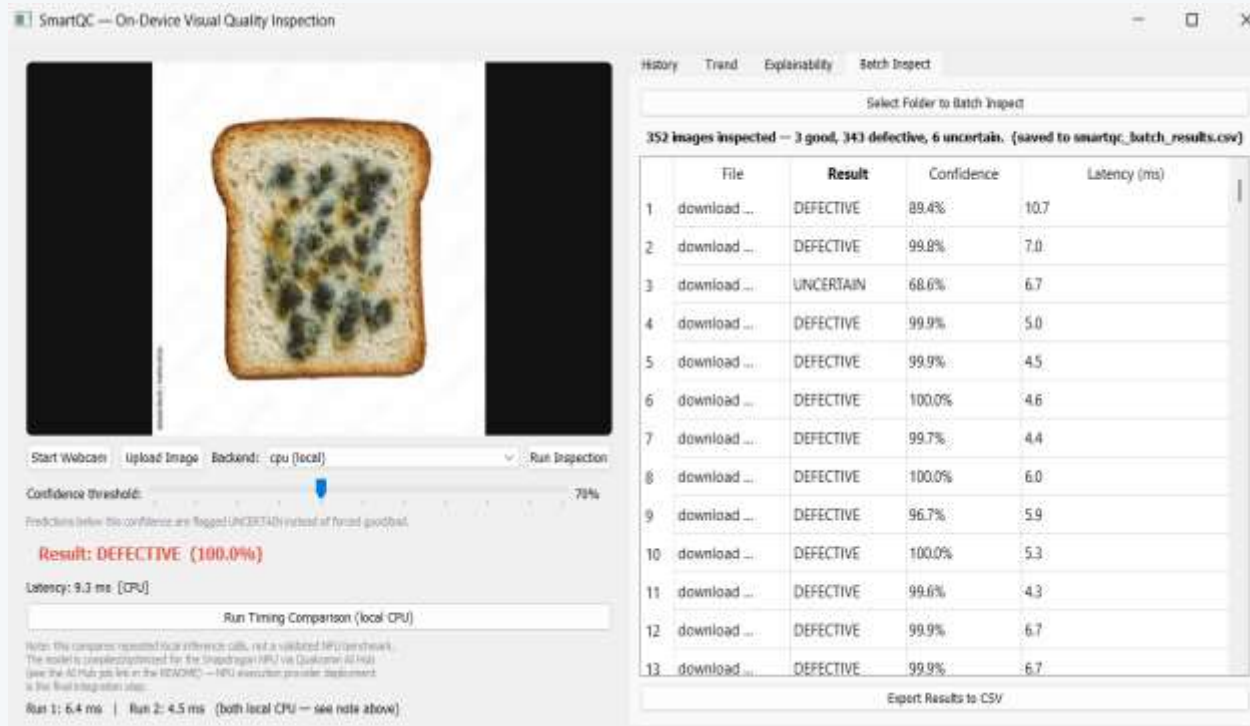
| History             |                   | Trend  |     |
|---------------------|-------------------|--------|-----|
| 2026-09-19 22:14:36 | defective (98.4%) | 4.3 ms | cpu |
| 2026-09-19 22:14:35 | defective (98.4%) | 4.5 ms | cpu |
| 2026-09-19 22:14:35 | defective (98.4%) | 3.8 ms | cpu |
| 2026-09-19 22:14:35 | defective (98.4%) | 3.9 ms | cpu |
| 2026-09-19 22:14:35 | defective (98.4%) | 8.0 ms | cpu |
| 2026-09-19 22:14:31 | defective (98.4%) | 3.8 ms | cpu |
| 2026-09-19 22:14:31 | defective (98.4%) | 4.1 ms | cpu |
| 2026-09-19 22:14:30 | defective (98.4%) | 4.0 ms | cpu |
| 2026-09-19 22:14:30 | defective (98.4%) | 4.5 ms | cpu |
| 2026-09-19 22:14:30 | defective (98.4%) | 4.5 ms | cpu |
| 2026-09-19 22:14:30 | defective (98.4%) | 3.9 ms | cpu |
| 2026-09-19 22:14:30 | defective (98.4%) | 4.3 ms | cpu |
| 2026-09-19 22:14:30 | defective (98.4%) | 4.3 ms | cpu |
| 2026-09-19 22:14:29 | defective (98.4%) | 4.8 ms | cpu |
| 2026-09-19 22:14:29 | defective (98.4%) | 4.9 ms | cpu |
| 2026-09-19 22:14:29 | defective (98.4%) | 4.3 ms | cpu |
| 2026-09-19 22:14:29 | defective (98.4%) | 4.3 ms | cpu |
| 2026-09-19 22:14:29 | defective (98.4%) | 4.2 ms | cpu |
| 2026-09-19 22:14:28 | defective (98.4%) | 5.6 ms | cpu |
| 2026-09-19 22:14:28 | defective (98.4%) | 4.2 ms | cpu |
| 2026-09-19 22:14:28 | defective (98.4%) | 4.8 ms | cpu |
| 2026-09-19 22:14:28 | defective (98.4%) | 4.6 ms | cpu |
| 2026-09-19 22:14:28 | defective (98.4%) | 4.4 ms | cpu |
| 2026-09-19 22:14:28 | defective (98.4%) | 4.4 ms | cpu |

DEFECTIVE — 98.4% confidence

## REAL-WORLD VALIDATION

# Batch-tested against 352 uncured images

Beyond the curated 129-image validation set, SmartQC's batch mode was run against a larger, unsorted folder of 352 real-world mold-bread images — a messier, more realistic test than a clean benchmark.



352

Images batch-processed

343

Correctly flagged  
DEFECTIVE

6

Flagged UNCERTAIN  
(below threshold)

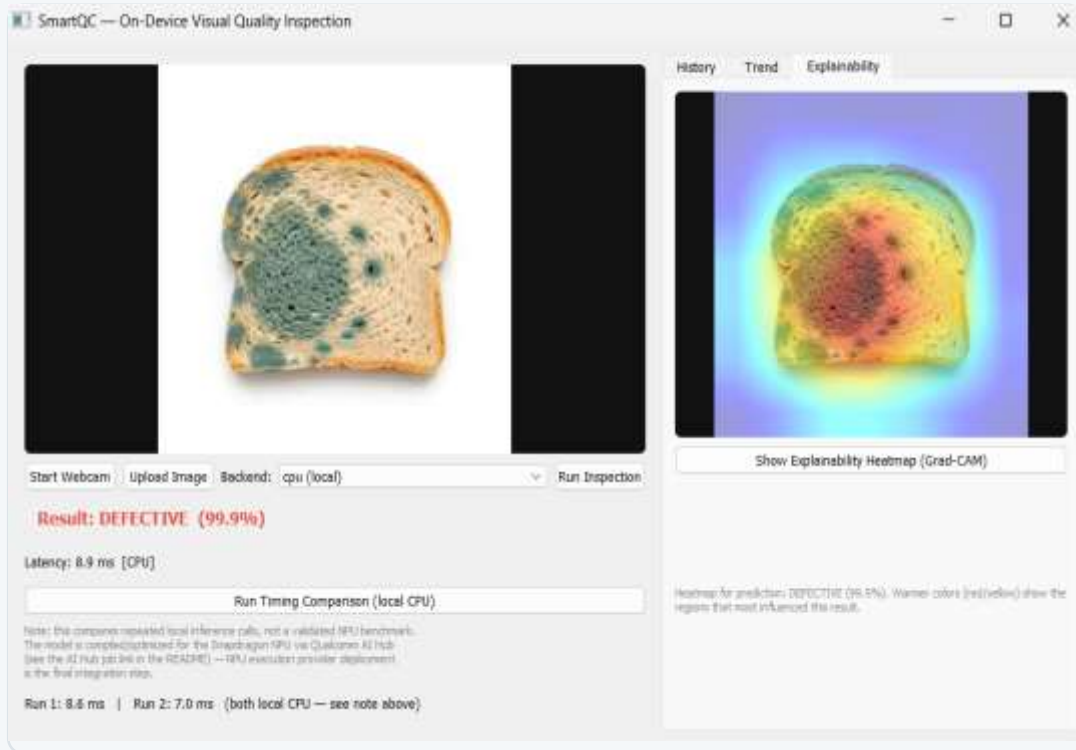
3

Misclassified (edge cases)

97.4% correct on an uncured real-world batch — and the confidence-threshold feature caught the ambiguous cases instead of silently guessing wrong.

## BEYOND CLASSIFICATION

# Explainability: showing WHY, not just what



### Grad-CAM heatmap

Highlights exactly which pixels drove the model's decision — warmer colors (red/yellow) show the highest-influence regions.

### Builds operator trust

A QC operator can visually verify the model is looking at the actual defect, not an unrelated artifact.

### Runs on the trained model directly

Computed via PyTorch alongside the ONNX/AI-Hub inference path — an explainability layer, not a replacement for it.

WHAT'S NEXT

# Roadmap

1

## **Live QNN NPU deployment**

Wire up the QNN Execution Provider on physical Snapdragon hardware for measured on-device NPU latency.

2

## **Multi-product support**

Extend beyond bread to additional inspection domains (produce, packaging, components).

3

## **Multi-class defects**

Move beyond good/bad to specific defect types — mold, staleness, physical damage.

4

## **Fleet / multi-station rollout**

Central dashboard aggregating results across multiple inspection stations.

# SNAPDRAGON® AI LAB BUILD & PRESENT CHALLENGE

## SmartQC

On-device inspection, privately and instantly, on Snapdragon.

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[Demo](#) | [GitHub](#) | [Linkedin](#)

# Thank You